Solution Requirements

Date	5 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Define the requirements of the **Credit Card Approval Prediction System** to specify what the system should do and how well it should perform.

- **Functional Requirements (FR):** Describe the features of the system, such as collecting applicant details, predicting credit card approval, and displaying the result.
- **Non-Functional Requirements (NFR):** Describe the quality of the system, including performance, security, reliability, and usability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Applicant Information Management	The system allows users to enter applicant details through a web-based form for credit card approval prediction.	High
2	Machine Learning Prediction	The system processes applicant information using a trained machine learning model and predicts whether the applicant is a Good Customer or Bad Customer.	High

3	External interfaces	The system interacts with the Flask web application, trained machine learning model, SQLite database, and visualization components.	High
4	Transactions processing	The system accepts applicant details, preprocesses the data, performs prediction, and displays the approval or rejection result	High
5	Reporting & Analytics	The system displays prediction results, confidence score, prediction history, dashboard statistics, and model performance reports.	High
6	Business Rules	Credit card approval prediction is based on applicant information such as income, employment, education, family status, housing type, and other input features.	High
7	Data Management	The system stores applicant prediction history, prediction confidence, date, and time in the SQLite database for future reference.	High
8	Input Validation	The system validates user inputs before processing the prediction to ensure accurate and reliable results.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate prediction results quickly.	Prediction displayed within **2 seconds**)
2	Scalability	The system should support increasing numbers of applicant records and future feature enhancements.	Supports database growth and modular expansion
3	Security & Data Privacy	Applicant prediction records should be securely stored and protected from unauthorized modification.	Secure SQLite storage and input validation
4	Reliability & Availability	The application should provide consistent prediction results whenever users access the system.	99% availability during normal operation
5	Usability & Accessibility	The web interface should be simple, responsive, and easy to use on different screen sizes.	User-friendly interface with responsive design

6	Maintainability	The system should be easy to maintain and update with new machine learning models.	Modular, reusable, and well-documented code